

Servo motor

DO

OUTPUT

MOVEMENT

What does it do?

A positional servo moves its arm, called a horn, to a chosen position and holds it there.

Short pulses on its signal wire set the position. The motor and position control are built into the case.

Use it for steering, pointing or opening a flap. A continuous-rotation servo controls speed instead; it cannot use this example to choose an angle.

Uses

Radio-controlled steering

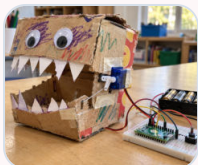


DETECT
Read steering
command

DECIDE
Which way
should it
turn?

DO
Set the
steering
position

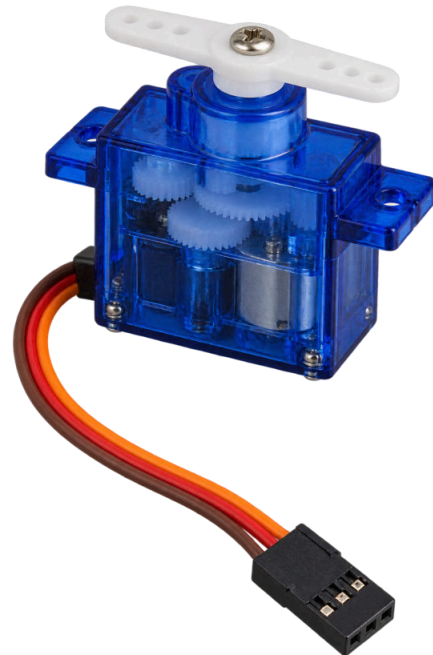
Cardboard chomping monster



DETECT
Read feeding
button

DECIDE
Is the monster
hungry?

DO
Open its
cardboard jaw



Horn
Attach a
light
moving
part

Signal
Usually
orange

Power
Usually
red;
separate
supply

Ground
Brown or
black; join
grounds

Code

● Raspberry Pi

```
from gpiozero import Servo
from time import sleep
```

```
servo = Servo(16)
servo.mid()
sleep(2)
```

● Raspberry Pi Pico

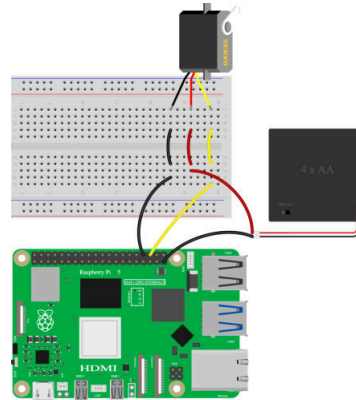
```
from picozero import Servo
from time import sleep
```

```
servo = Servo(15)
servo.mid()
sleep(2)
```

Test unloaded: the horn centres. Try `min()` or `max()` instead of `mid()`. Travel varies; stop if it binds or buzzes. Check lead order.

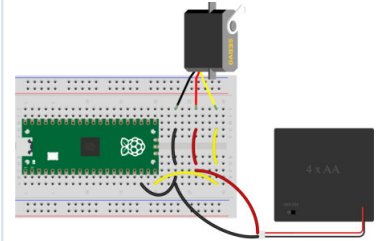
Wiring

RASPBERRY PI 5



GPIO16 (pin 36) → signal; **GND** pins
34/39 → servo/pack -. **4xAA NiMH (4.8
V)** pack + → servo red.

PICO



GP15 → signal; **GND** (pin 18) → servo/pack -. Use **4×AA NiMH (4.8 V)**; USB powers the board. Keep pack + off board power pins.